Bank Marketing: End-to-End Supervised Learning Pipeline

Course	MSc in Data Science and AI: Machine Learning
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1. Objective

Assignment evaluates ability to construct complete supervised learning workflow, encompassing data preprocessing, training & evaluation of various classification models including Decision Trees, Rule-Based models, kNN & Ensemble Learning methods.

2. Dataset Description | https://www.kaggle.com/datasets/adityamhaske/bank-marketing-dataset

Project utilizes UCI Bank Marketing Dataset, publicly available on Kaggle, dataset contains information from direct marketing campaigns (phone calls) of banking institution. Primary goal is to predict whether bank client will subscribe to a term deposit ('Yes') or not ('No') after being contacted.

- **Total Records:** 45,211 customers
- **Features:** 17 attributes, cover client demographics, financial details & past campaign outcomes.
- **Target Variable:** `y` (renamed to `deposit`), subscription to term deposit (binary: 'yes'/'no').

Key Features

- **Client Data:** age, job, marital, education, default (credit in default), balance (average yearly balance), housing (housing loan), loan (personal loan).
- **Last Contact Information:** contact` (communication type), day (last contact day of month), `month` (last contact month), duration (last contact duration in seconds).
- **Other Attributes:** campaign` (number of contacts during this campaign), pdays (days since last contact from previous campaign, -1 if never contacted), previous (number of contacts before this campaign), poutcome (outcome of previous campaign).

3. Data Quality & Preprocessing

3.1. Missing Values, Duplicates & Noisy/Outlier Data

a. Missing Values: Initial checks confirmed absence of NaN values, empty strings or whitespace entries. Thus, no imputation or removal of missing data was required.
b. Duplicates: No duplicate records were identified. Each entry corresponds to unique customer interaction, ensuring strong data integrity for analysis & model training.
c. Noisy/Outlier Data:

- Numerical features balance, duration, campaign & previous exhibited outliers, were genuine extreme cases (e.g., high balances or long call durations) rather than errors. Notably, longer call durations often correlate with successful conversions. Outliers retained to preserve patterns.
- Categorical variables unknown (in job, education) & dec (in month), preserved & handled during encoding, as they reflect valid but infrequent or missing information.

3.2. Attribute Types

a. Nominal Attributes: Categorical attributes without any inherent order or ranking.

• job: (management, technician, blue-collar, etc) • marital: (married, single, divorced) • default: (no, yes) • housing: (yes, no) • loan: (no, yes)	• contact: (unknown, cellular, telephone') • month: (jan, feb, ..., dec) • poutcome: (unknown, failure, other, success') • `deposit (target variable): (no, yes)

b. Ordinal Attributes: Categorical attributes with a meaningful order or ranking.

- education: (primary, secondary, tertiary, unknown) implies hierarchy of educational attainment.

c. Numeric Attributes: Attributes with numerical values subject to arithmetic operations.

i. **Discrete Numeric:** Can only take finite or countably infinite number of values (often integers).

• age • day • campaign	• pdays (with -1 indicating no prior contact) • previous

ii. **Continuous Numeric:** Can take any value within given range (often decimals/fractions).

- Balance
- duration

3.3. Data Preprocessing

3.3.1. Encode Categorical Variables

- **Binary Categorical Features:** deposit (target), loan, default & housing were converted into numerical representations (0s & 1s).
  - For deposit, yes mapped to 1, no to 0.
  - For loan, default & housing, no was mapped to 1 & yes to 0, signifying absence of negative financial condition as a 'positive' state.
- **Multi-Category Features (One-Hot Encoding):** marital, education, job, contact, month & poutcome were one-hot encoded i.e. each category into new binary column. To avoid multicollinearity (dummy variable trap), one category from each feature was dropped (e.g., marital_divorced, education_unknown, job_unknown, contact_unknown, month_dec, poutcome_other).

3.3.2. Feature Scaling (Standardization)

Numerical features scaled using `StandardScaler` after train-test split to prevent data leakage, for transforming data to have mean of 0 & standard deviation of 1, crucial for algorithms sensitive to feature magnitudes (e.g., kNN, Logistic Regression & tree-based methods).

Formula used **Z = (X - U) / S**, where Z is scaled value, X is original value, U is mean, and S is standard deviation.

3.3.3. Dataset Splitting

Dataset split into training & testing sets with **80:20** ratio, ensuring model is trained on substantial portion of data & evaluated on unseen data to provide unbiased assessment of generalization capabilities.

- **Training Features (X_train):** 36,168 samples with 42 features.
- **Testing Features (X_test):** 9,043 samples with 42 features.
- **Training Labels (y_train):** 36,168 labels.
- **Testing Labels (y_test):** 9,043 labels.

3.4. Handling Class Imbalance (SMOTE)

- Target variable `deposit` was imbalanced, with significant majority of 'no' subscriptions.
- To address, Synthetic Minority Over-sampling Technique (SMOTE) applied to training data.
- Minority class (`deposit = 'yes'`) oversampled to 4:1 ratio relative to majority class, to prevent models biased towards majority class, which aids to improve their ability for correctly identifying potential subscribers, critical for bank's marketing objectives.

4. Model Training & Evaluation

4.1. Baseline Models

Logistic Regression	Naive Bayes (GaussianNB)
• **Accuracy:** 0.8795	• **Accuracy:** 0.8474
• **Precision:** 0.4867	• **Precision:** 0.3894
• **Recall:** 0.6952	• **Recall:** 0.5533
• **F1 Score:** 0.5725	• **F1 Score:** 0.4571
• **ROC-AUC:** 0.9084	• **ROC-AUC:** 0.8066
• **Confusion Matrix:** [[7223 770] [320 730]]	• **Confusion Matrix** [[7082 911] [469 581]]

Interpretation

- Logistic Regression showed good accuracy & strong recall, identifying potential subscribers; moderate precision indicates more false positives, causing wasted marketing effort.
- Naive Bayes performed lower than Logistic Regression, with reduced precision & recall, making it less effective in balancing true positive detection & false positive minimization.

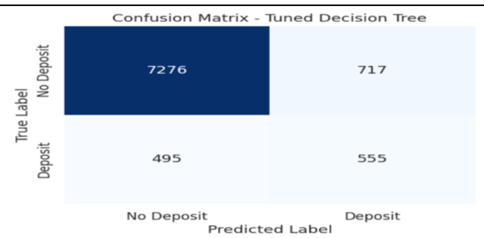
4.2. Decision Tree Learning & Interpretation

Trained & optimized using GridSearchCV to tune its hyperparameters.

Hyperparameter Tuning:

- **Parameters Tuned:** `max_depth`, `min_samples_split`, `min_samples_leaf`.
- **Best Parameters Found:** `{'max_depth': None, 'min_samples_leaf': 1, 'min_samples_split': 2}`
- **Best F1 Score (from GridSearchCV):** 0.8213

Evaluation Metrics

• **Accuracy:** 0.8660 • **Precision:** 0.4363 • **Recall:** 0.5286 • **F1 Score:** 0.4780 • **ROC-AUC:** 0.7194	

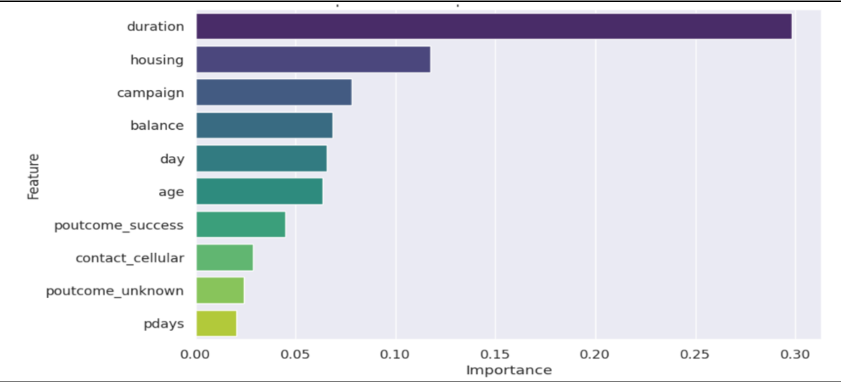
Interpretation: Achieved moderate performance, while its accuracy is acceptable, F1-score & AUC are lower than Logistic Regression, suggesting it may struggle with capturing nuanced relationships in data as effectively as more complex models.

4.2.1. Feature Importance

Most influential features for Decision Tree model's predictions.

Interpretation: duration of last contact is by far most crucial feature, highlighting its significant impact on subscription decisions. Other features housing (having a housing loan), campaign (number of contacts), balance, poutcome_succes & age also play important roles.

- duration: 0.2985
- housing: 0.1175
- campaign : 0.0783
- balance: 0.0685
- day: 0.0658
- age: 0.0635
- poutcome_success: 0.0452
- contact_cellular: 0.0289
- poutcome_unknown: 0.0244
- pdays: 0.0203



4.2.2. Decision Path Interpretation

a. Path 1 (Predicting 'No Deposit')

```
|--- duration <= -0.21 (e.g., short contact duration)
|   |--- poutcome_success <= -0.16 (e.g., previous campaign was not a success)
|   |   |--- housing <= -0.89 (e.g., client has a housing loan)
|   |   |   |--- month_may >  0.65 (e.g., contact occurred in May)
|   |   |   |   |--- age >  -0.42 (e.g., client is not very young)
|   |   |   |   |   |--- class: 0 (No Deposit)
```

Illustrates combination of short contact, previous campaign failure, presence of housing loan, contact in Ma & non-youthful age collectively leads to prediction of 'No Deposit'.

b. Path 2 (Predicting 'Deposit')

```
|--- duration >  0.83 (e.g., notably long contact duration)
|   |   |--- duration <= 1.53 (e.g., contact duration still within an effective high range)
|   |   |   |--- housing >  -0.89 (e.g., client does not have a housing loan)
|   |   |   |   |--- housing <= 1.11 (e.g., confirming no housing loan)
|   |   |   |   |   |--- class: 1 (Deposit)
```

Indicates significantly long contact duration combined with absence of housing loan strongly predicts 'Deposit' outcome.

4.3. Rule-Based Classification

Extracted rules directly from trained Decision Tree, to leverage inherent interpretability of decision trees to create set of explicit if-then rules.

4.3.1. Extracted Rules

Rules	Key Conditions	Prediction
1	Short duration + No previous success + Housing loan + May contact + Not very young	No Deposit (0)
2	Long duration (high range) + No housing loan	Deposit (1)
3	Very short duration + Previous success + No/old prior contact	No Deposit (0)
4	Short duration + No previous success + No housing loan	Deposit (1)
5	Moderate/long duration + No housing loan + Positive previous signals	Deposit (1)

4.3.2. How Rules Affect Interpretability vs. Performance

a. Interpretability: Rule-based models from Decision Trees are highly interpretable, with clear if-then logic for predictions. Such transparency aids stakeholders understand decisions, build trust enabling actionable insights.

b. Performance: Decision Trees may achieve lower accuracy than complex ensemble models, as they struggle with intricate non-linear patterns which reflects trade-off where accuracy is sacrificed for interpretability, preferred in explainable scenarios.

4.4. kNN (Lazy Learning)

K-Nearest Neighbors (kNN) classifier was trained, with `GridSearchCV` to experiment with different `k` values and distance metrics.

Hyperparameter Tuning	Evaluation Metrics
• **Parameters Tuned:** n_neighbors (k values: [1, 3, 5, 7, 9]), metric (euclidean, manhattan). • **Best Parameters Found:** {metric: manhattan, n_neighbors: 1}` • **Best F1 Score (from GridSearchCV):** 0.9196	• **Accuracy:** 0.8726 • **Precision:** 0.4479 • **Recall:** 0.4171 • **F1 Score:** 0.4320 • **ROC-AUC:** 0.6748 • **Confusion Matrix:** [[7453 540] [612 438]]

Interpretation: Despite finding parameters during tuning, showed relatively lower recall & F1-score on test set compared to Logistic Regression & its AUC was also modest. Suggests that while it tries to capture local patterns, it might be sensitive to specific data distribution on test set, even after scaling.

4.4.1. Why Scaling is Critical for kNN

Feature scaling is critical for kNN as it relies on distance calculations; without scaling, features with larger magnitudes dominate & bias results. Scaling (e.g., using StandardScaler) ensures all features contribute proportionally, leading to accurate similarity measurement and better predictions.

4.4.2. How k Affects Bias/Variance in kNN

1. Small k (e.g., 1) leads to high variance & low bias, makes model sensitive to noise & prone to overfitting with complex decision boundaries.
2. Large k (e.g., 9+) results in low variance & high bias, produces smoother boundaries but risking underfitting by oversimplifying patterns.
3. Optimal k balances trade-off for better generalization, determined using cross-validation (e.g., GridSearchCV).

4.5. Ensemble Learning

Random Forest	XGBoost	AdaBoost (Adaptive Boosting)
• **Accuracy:** 0.8982 • **Precision:** 0.5776 • **Recall:** 0.4571 • **F1 Score:** 0.5104 • **ROC-AUC:** 0.9037 • **Confusion Matrix:** [[7642 351] [570 480]]	• **Accuracy:** 0.9048 • **Precision:** 0.5961 • **Recall:** 0.5581 • **F1 Score:** 0.5765 • **ROC-AUC:** 0.9310 • **Confusion Matrix** [[7596 397] [464 586]]	• **Accuracy:** 0.8742 • **Precision:** 0.4671 • **Recall:** 0.5952 • **F1 Score:** 0.5235 • **ROC-AUC:** 0.8859 • **Confusion Matrix** [[7280 713] [425 625]]

Interpretation:

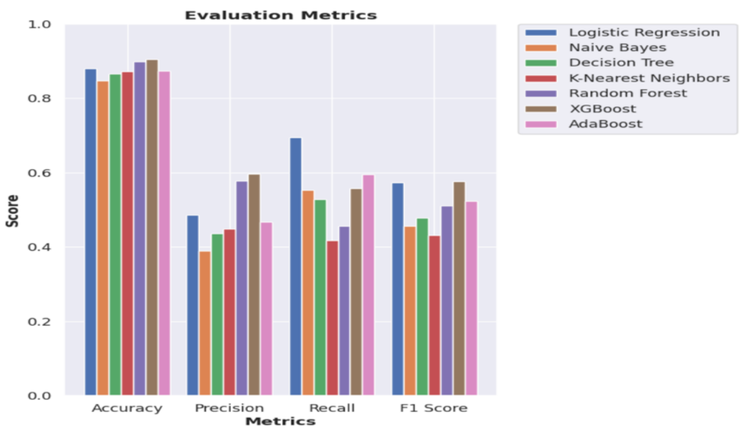
a. For **Decision trees**, strong accuracy & good AUC indicate effective class separation; higher precision but moderate recall suggests conservative approach in predicting positive cases
b. **XGBoost** achieved best overall performance with highest F1-score & AUC, strong discriminative power & balanced precision-recall, making it highly robust.

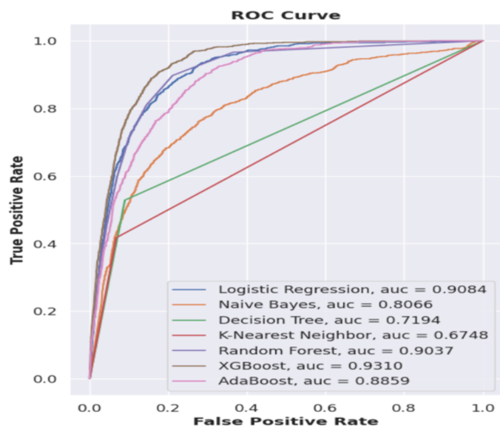
c. **AdaBoost** performed reasonably well with balanced precision, recall & decent AUC, improving over Decision Trees but slightly lagging behind Random Forest & XGBoost.

5. Final Comparison & Recommendation

5.1. Comparison

	Model	Accuracy	Precision	Recall	F1 Score	AUC
0	Logistic Regression	0.8795	0.4867	0.6952	0.5725	0.9084
1	Naive Bayes	0.8474	0.3894	0.5533	0.4571	0.8066
2	Decision Tree	0.8660	0.4363	0.5286	0.4780	0.7194
3	K-Nearest Neighbors	0.8726	0.4479	0.4171	0.4320	0.6748
4	Random Forest	0.8982	0.5776	0.4571	0.5104	0.9037
5	XGBoost	0.9048	0.5961	0.5581	0.5765	0.9310
6	AdaBoost	0.8742	0.4671	0.5952	0.5235	0.8859





5.2. Which model performed best and why?

XGBoost emerged as best-performing model across key metrics, for predicting term deposit subscriptions.

- Accuracy (0.9048): Best overall classification of both subscribers & non-subscribers.
- F1-Score (0.5765): Balance b/w precision & recall, identifying positives with fewer errors.
- AUC (0.9310): Excellent ability to distinguish between classes across thresholds.

5.3. Which model is most interpretable and why?

Decision Tree is most interpretable model due to its clear, rule-based decision paths.

- Clear Decision Paths: Uses intuitive if-then rules, making predictions easy to understand.
- Transparency: Provides direct reasoning behind predictions, enabling actionable insights.
- Contextual Feature Importance: Shows how features influence decisions at each step.

5.4. If model was deployed in real life, what risks exist?

1. **Over-reliance on duration**: Not available before calls, leading to poor pre-targeting decisions.
2. **Data Drift**: Changing customer behavior can degrade model performance over time.
3. **Bias & Fairness**: Historical bias may lead to unfair or skewed predictions.
4. **False Positives/Negatives**: Causes wasted marketing effort or missed opportunities.
5. **Low Explainability**: XGBoost acts as a black box, reducing trust & debuggability.
6. **Privacy & Customer Impact**: Targeted outreach may raise privacy concerns or cause annoyance.
7. **Mitigation**: Continuous monitoring, retraining, A/B testing & feedback loops essential.